

Identification and Tracking of Rossby Wave Packets

A Thesis

submitted to

Indian Institute of Science Education and Research Pune

in partial fulfillment of the requirements for the

BS-MS Dual Degree Programme

by

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May, 2022

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Certificate

This is to certify that this dissertation entitled "Identification and Tracking of Rossby Wave Packets" towards the partial fulfilment of the BS-MS dual degree programme at the Indian Institute of Science Education and Research, Pune represents study/work carried out by Malavika Biju at Indian Institute of Science Education and Research under the supervision of Dr. Joy Merwin Monteiro, Assistant Professor, Department of ECS, during the academic year 2021-2022.

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This thesis is dedicated to my parents.

Declaration

I hereby declare that the matter embodied in the report entitled "Identification and Tracking of Rossby Wave Packets", are the results of the work carried out by me at the Department of ECS, Indian Institute of Science Education and Research, Pune, under the supervision of Dr. Joy Merwin Monteiro and the same has not been submitted elsewhere for any other degree.

Malavika Biju

Acknowledgments

First and foremost, I thank my supervisor, Dr. Joy Monteiro. His guidance and invaluable support have helped me conduct original work in the field of climate sciences, to which I was completely new. Special thanks to him for forgiving all my amateur mistakes. I also thank my co-supervisor, Dr. Vijay Natarajan, whose suggestions and guidance have immensely helped me during the second phase of my work. The advice given by my TAC member Dr. Suhas Ettammal has helped me improve my domain knowledge.

My friends and family have been a constant support throughout my BS-MS journey at IISER, Pune. I thank my friends, Yashi Jain and Shweta Singh, for always being up for discussions on mathematics, broader goals, and opportunities in academia and otherwise. I thank my friend Suhail Odungat for being a pillar of mental support throughout the pandemic and other difficult times.

Finally, I thank INSPIRE (Innovation in Science Pursuit for Inspired Research) for funding my study at IISER, Pune

Abstract

Rossby Wave Packets (RWPs) are regions of high amplitude meridional winds in the upper troposphere that acts as precursors to extreme weather phenomena. Due to this, the identification and tracking of RWPs is an important field of research. We have implemented modifications to an identification algorithm that uses the underlying topology of the meridional wind field and techniques of computational geometry to identify RWPs from gridded data. These modifications have helped improve efficiency and reduce computational complexity. We have also implemented a feature-based tracking approach to track the evolution of RWPs across time. We recognize regions enclosing the identified RWPs as features. These features are then stored in the form of quadrees for each time step which optimises storage and facilitates easy computation of unions and sizes of features. We then compute the overlap of features in consecutive time steps using their respective quadtree representations and formulate a weight that determines the extent of each overlap. Finally, we construct a tracking graph consisting of nodes representing features in each timestep and weighted edges representing the extent of overlap between these features. This representation allows for easy analysis of RWP tracks and contains information about splits and merges of RWP tracks.

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Chapter 1

Introduction

1.1 Background

A Rossby Wave Packet (RWP) is a localized region of high amplitude meridional flow in the upper troposphere [SCC14a]. Its central characteristic is the propagation in the zonal direction. It also has a group velocity greater than the phase velocity, which leads to the transfer of wave energy from one individual trough or ridge to its downstream neighbor, a process called “downstream development” [WRCM18].

There are many different meteorological processes that can cause an RWP. Some of these include diabatic heating in the mid-lower troposphere due to pre-existing synoptic disturbances such as extratropical cyclones and recycling from earlier waves in the jet-stream waveguide. The strength and localization of the large-scale background potential vorticity (PV) gradient, which functions as a waveguide, is largely responsible for RWP propagation and extension. Such coherent, long RWPs are more common in the Southern Hemisphere. In the Northern Hemisphere, RWPs are often shorter and more difficult to trace due to the existence of broad, baroclinically unfavourable continental expanses. [GV]

There has been an increasing interest in the RWPs along the midlatitude waveguide. The major reason for this is that these RWPs often act as precursors to extreme weather events in the midlatitudes. Extreme weather events are events based on local recorded weather

history and lie in the tail of the respective local distribution. Hence a better understanding of RWP dynamics would help in improving the prediction of these extremes. RWPs have also been linked to large-scale regime changes and downstream propagation of model errors and uncertainties[\[Hak05\]](#)[\[ABKC13\]](#).

1.2 Related Work

The importance of RWPs has motivated many studies to provide a precise algorithmic description to track and identify RWPs. In this section, we discuss the diagnostic frameworks that have been developed earlier for the study of RWP dynamics.

1.2.1 Hovmöller Diagrams

The Hovmöller diagram, designed in 1948 by Ernst Hovmöller, is a diagram in which the latitudinal average of a variable from the upper troposphere (such as geopotential, meridional wind, meridional wind squared, etc.) is plotted in a longitude-time map [\[WRCM18\]](#). This method condenses the spatiotemporal information and makes use of the excellent pattern recognition skills of the human brain to track individual RWPs across time. The main limitation of this method is that since it uses a latitudinal average, we cannot obtain information about the latitudinal propagation of the RWPs. This would limit the application of these diagrams to long time series as RWPs may shift latitudinally with time (Eg: as a function of seasons). There have been many modifications to overcome the shortcomings of the classical Hovmöller diagrams. For example, [\[GDJ⁺11\]](#) proposes the use of a latitudinal weighting function that favors those latitudes where the longitudinal variance of the variable is large instead of averaging over a fixed latitude band. However, these methods cannot detect the two-dimensional horizontal structure of the wave packets and can lead to erroneous results when multiple RWPs exist in close proximity at different latitudes.

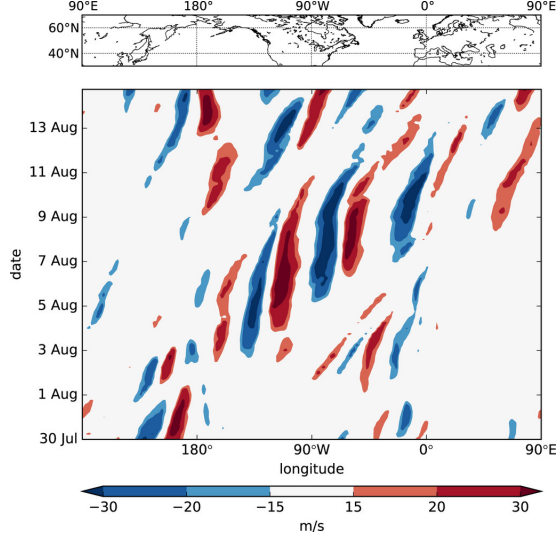


Figure 1.1: The Hovmöller diagram shows the 250-hPa meridional wind (the colour bar is in m/s) during the August 2002 time period, using data averaged between latitudes 40° and 60°N. The dates are July 30 – August 15, 2002. The map on top of the illustration aids spatial navigation.[[WRCM18](#)]

1.2.2 Envelope Reconstruction Methods

We require a more objective and automated approach than the previous for identification and tracking of RWPs from large datasets. In this direction envelope reconstruction methods such as complex demodulation, Hilbert transform based methods have been developed.

- Complex demodulation: This technique assumes a spatially dependent scalar atmospheric quantity, $v(x)$ can be written in the form of $v(x) = \text{Re}[A(x)\exp(ikx)]$ where k is a supposed carrier wavenumber. Then we demodulate by multiplying $v(x)$ with $\exp(-ikx)$ and finally extract the absolute value of the wave packet envelope, $|A(x)|$ using low-pass filtering to remove higher wavenumber components. Though we generally obtain reasonable results when carrier wavenumber is chosen from a plausible range of values, this method would yield incorrect results when multiple wave packets of different carrier wavenumbers coexist in the same latitude[[ZSP+03](#)].
- Hilbert transform based technique: [[ZSP+03](#)] suggested an alternative technique to extract the envelope of wave packets that is not affected by the aforementioned problem. It involves Hilbert transform along a latitudinal circle. Unlike complex demodulation, this method does not require a prespecified carrier wavenumber.

However, both these envelope reconstruction methods implicitly assume that RWPs are purely oriented in the zonal direction and are almost plane wave packets which are not always true. Furthermore, they also assume that the wave packets must be small amplitude to allow linearization. However, observed RWPs are often large amplitude and often display non linear behaviors such as wave-breaking and cutoff-formation, which contradicts the above assumption. To overcome these shortcomings [GFW18] introduced Local finite-amplitude wave activity (LWA), a diagnostic method to identify RWPs and quantify their amplitude. The above mentioned algorithms provide a description of RWPs at the expense of losing phase information. However, the phase information is often important for characterizing the local synoptic situation at a location.

Combining the identification step with a suitable tracking algorithm would help us to effectively extract information about RWPs from large datasets.[SCC14b] describes feature-based tracking techniques to track RWPs. They use two objective approaches for tracking: a point-based cost optimisation method and a hybrid method involving both point-based and object-based methods. Point-based tracking method involves identifying the significant relative maxima and minima points in the atmospheric data as objects to be tracked. On the other hand, object-based tracking method considers the entire enclosed shape of a wave packet as the object to be tracked and uses approaches such as statistical similarities, overlap or object proximity between two time steps for tracking. The hybrid tracking approach proposed in [SCC14b] uses both the local extrema of RWP amplitudes and the overall structure of the wave packet.

1.3 Topological Methods in Climate Visualisation

Topology-based methods have proved to be a versatile approach to analysing scientific data. There are numerous topological models that can be used for the analysis of scalar fields. These include critical points and persistent homology, merge trees, contour trees, Reeb graphs, Morse Smale Complexes, and extremum graphs. We discuss a few important topological concepts below:

- **Level Sets:** Level sets are important topological descriptors of a dataset. The level set

of a function $f : \Omega \rightarrow \mathbb{R}$ is its preimage for some value ν :

$$f^{-1}(\nu) = \{x \in \Omega | f(x) = \nu\}$$

This preimage would consist of multiple connected components which are called isocontours. We define the concept called Reeb Graph using the idea of isocontours. Similar to level sets, we can also define sublevel and superlevel sets of a function. The sublevel set of a function $f : \Omega \rightarrow \mathbb{R}$ for a value ν is given by $\{x \in \Omega | f(x) \leq \nu\}$. Similarly, the superlevel set of f for a value ν is given by $\{x \in \Omega | f(x) \geq \nu\}$.

- **Morse Theory:** It is a mathematical theory that enables the study of the topology of a domain using functions that are defined on that domain. To understand Morse Theory, let us first define a Morse function. Let $\mathbb{M}$ be a smooth manifold and $f : \mathbb{M} \rightarrow \mathbb{R}$ a smooth function. The points in $\mathbb{M}$ where the partial derivatives of f vanish are called critical points and their images are called critical values. A critical point $x \in \mathbb{M}$ is said to be non-degenerate if the matrix of second partial derivatives (Hessian matrix) of f is non-singular. If all the critical values of f are non-degenerate and have distinct function values, f is a Morse function. A main result of Morse theory is that the set of Morse functions is an open dense subset of the space of smooth functions. In other words, almost all smooth functions are Morse functions. Two fundamental theorems of Morse theory is about the study of changes of the sublevel sets of a function with respect to its critical points. Briefly, these theorems indicate if and when the topology of the sublevel sets $\mathbb{M}_t$ vary as t changes.

However, most of the time we are not given a smooth manifold $\mathbb{M}$. We are given a point cloud sample of $\mathbb{M}$. Therefore, in these cases, we use combinatorial structures (simplicial complexes) on the sample point cloud as an approximation of the underlying manifold $\mathbb{M}$. Examples of simplicial complexes are triangulations, lattices, quadrangular meshes etc. In this case, we define a piece-wise linear function on this complex and use ideas from Morse Theory on this piece-wise linear approximation[YMS⁺21][RKG⁺11].

- **Merge Graph:** A merge graph (tree) is a tool to study the changes in connectivity in the sublevel sets of a Morse function $f : \mathbb{M} \rightarrow \mathbb{R}$ (where $\mathbb{M}$ is connected). We define two points $x, y \in \mathbb{M}$ to be equivalent (w.r.t f) if x, y belong to the same connected component of $\mathbb{M}_t$ for some $t \in \mathbb{R}$. A connected component is represented by a node in the tree and whenever a distinct connected component appears as t varies, a new

node is added to the tree. Therefore, this tree helps us to track critical points as well as record the evolution of the connectivity of $\mathbb{M}_t$ as t changes. Further discussion on computation of merge trees for piece-wise linear functions can be found in [HLH⁺16].

- **Reeb Graph:** Reeb graphs, on the other hand, are used to study the changes in connectivity in the level sets of a Morse function $f : \mathbb{M} \rightarrow \mathbb{R}$ (where $\mathbb{M}$ is connected). We define two points $x, y \in \mathbb{M}$ to be equivalent (w.r.t f) if $f(x) = f(y) = t$ and x, y belong to the same connected component of $f^{-1}(t)$ for some $t \in \mathbb{R}$. A Reeb graph is a quotient space on $\mathbb{M} / \sim$, where the equivalence relation is as described above. If the domain of f is simply connected, then the Reeb graph would have a tree structure and is called a contour tree. Similar to merge trees, Reeb graphs help us to record the evolution of the connectivity of the level sets of f as t changes. Further discussion on computation of Reeb graphs and contour trees can be found in [HLH⁺16].

Topological techniques have been widely used in climate sciences for feature tracking. In most cases, feature-tracking involves the identification of topological features in each dataset and matching them across time using a correspondence mechanism. [RKWH12] tracks critical points of 2D time dependent scalar fields using a combinatorial algorithm. This is done by converting the scalar field at each time step into a 3D combinatorial vector field and using a path search for the identification and tracking of critical points. This method is robust and computationally fast and has been successfully used for vortex tracking by [KHNH12].

[VMN⁺19] uses a topological pipeline for the identification and tracking of cyclones. This method first identifies cyclone centers within each time slice using local minima of the mean sea level pressure. To produce the final tracks, candidate tracks are calculated from an optical flow field, which is then clustered in a moving time window.

[DNN13] describes a method for exploration and tracking of cloud systems using infrared brightness temperatures. Merge trees are used for the computation and exploration of clouds. The threshold for clouds of interest is determined using a persistence diagram. Cloud movement is then tracked using an optical flow field computation. A tracking graph is then constructed using the optical flow between the computed clouds.

Chapter 2

Identification and Visualisation of RWPs

In this chapter, we discuss the formulation of the identification problem along with the details of the algorithm proposed in [\[PMN20\]](#) and the modifications made to the algorithm.

2.1 The Identification problem

As discussed earlier, Identification of RWPs using Fourier transform techniques results in the loss of phase information which is often important for characterizing RWPs. Hence [\[PMN20\]](#) proposes an algorithm for the identification of RWPs using the topology of the meridional wind field to obtain a concise description of RWPs in the form of a graph without using Fourier transforms. The nodes of this graph represent individual wave components in the form of maxima and minima clusters, and the edges represent spatial adjacency between the clusters. The maxima (minima) clusters consist of local maxima (minima) points of the wind field. This approach does not make any assumption about the wavelike nature of RWPs hence does not result in the loss of phase information. The graph representation also enables fast computation and recomputation of RWPs for different parameter values.

2.2 Data

All datasets are from [ECMWF](#) interim data downloaded in the form of NetCDF files. We use the 300 hPa meridional winds, v , with a spatial resolution of 0.75° in latitude and longitude space and a temporal resolution of 6 h. Data is available at each grid point, and we assume linear interpolation along axes for all computations. The algorithm does not assume a specific resolution or projection of the input data. We use the python library NetworkX to store and process graphs. The maxima and minima clusters, along with their associations, are stored as a list of nodes and edges using the NetworkX library.

2.3 Methodology

In this section, we provide the details of each step in the identification algorithm.

- Identifying the critical points: Critical point analysis is an important technique in topological data analysis. We also saw many examples in the previous chapter where critical point analysis has been successfully used in climate sciences. [\[PMN20\]](#) use a combinatorial approach for the computation of extrema instead of using gradients to find the critical points. The local extrema are located at points whose grid value is higher/lower than its neighbouring points. We are interested in capturing the cluster-like behavior of the extrema in the meridional wind field v . To avoid the detection of insignificant critical points, we apply a filter of 5m/s to remove all local maxima and minima below this value.

Figure 2.1 shows the initial local maxima identified in the northern hemisphere on 21 January 2007 before the 5m/s filter is applied. Similarly, figure 2.2 shows the initial local minima identified in the northern hemisphere on 21 January 2007 before this filter is applied.

- Computing maxima and minima clusters: The critical points located in the previous step has to be divided into maxima and minima clusters. Each cluster represents a potential component of an RWP. The maxima clusters are a set of proximal local maxima that lies on the same connected components of the superlevel set of 0. Similarly,

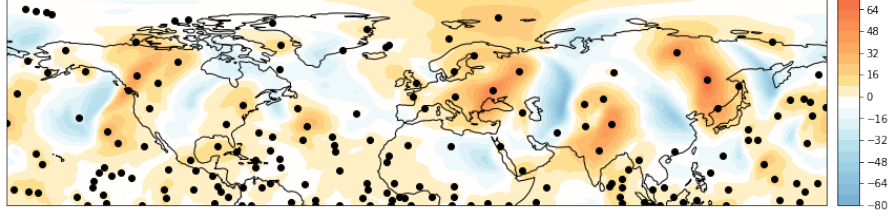


Figure 2.1: Local Maxima of the 300hPa meridional wind field on 21 Jan 2007 (0600 UTC)

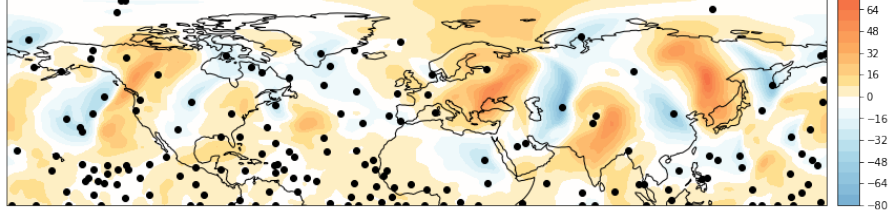


Figure 2.2: Local Minima of the 300hPa meridional wind field on 21 Jan 2007 (0600 UTC)

minima clusters are a set of proximal local minima that lies on the same connected components of the sublevel set of 0. For the computation of this partition, we use the affinity propagation algorithm[FD07] along with a similarity measure that takes both these requirements into account [FD07].

The affinity propagation algorithm works by recursively exchanging messages between the data points until a good set of stable clusters and corresponding representative member values emerge. The messages are based on the suitability of an input point to become an exemplar (representative point) for another point in the dataset. The algorithm takes as input the similarities between the data points, which denotes how well one data point can represent the other, and “preferences”. Preferences are the diagonal values of the similarity matrix, specified by the user to control the number of clusters. If a point has a higher diagonal value, then it has a higher chance of being picked as an exemplar. If at the beginning all points are equally likely to be exemplars, the diagonal similarity should be set to the same value for every point in the dataset. This value would influence the number of clusters. For example, taking the median of all similarities between pairs of points would yield a reasonable number of clusters, while taking the minimum of the similarities would result in a reduced number of clusters [FD07].

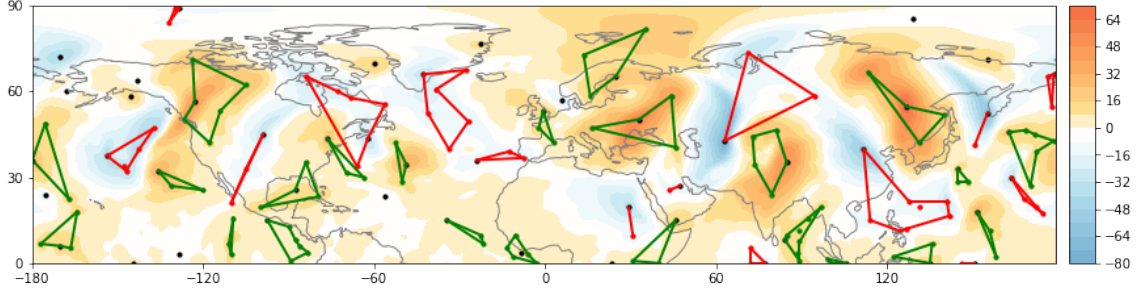


Figure 2.3: Clustering applied to 300hPa meridional wind field on 21 Jan 2007 (0600 UTC). Black points represent filtered critical points. Red shapes represent minima clusters while green shapes represent maxima clusters.

The details of the formulation of the similarity measure is given by [PMN20]. It depends on the length of the shortest path between two local maxima/minima points that lies on superlevel/sublevel sets of a given threshold value (2m/s and -2m/s in this case) and the minimum/maximum value along the straight line connecting two local maxima/minima.

The output of the clustering process is shown in the figure 2.3. The red shapes represent minima clusters and the green shapes represent maxima clusters. The clustering algorithm in this case uses 150 as the diagonal element in the similarity matrix. We can also see from the diagram that the zero-isocontour divides the wind field into separate regions. Therefore, each maxima and minima cluster has a particular region associated with it.

- Computing the association graph: The next step in the identification algorithm is to identify the relevant associations between maxima clusters and minima clusters to construct an association graph. This is an important step as the maxima clusters and minima clusters identified earlier together form a wave packet. To identify these pairwise associations, we look for evidence for a shared boundary between pairs of maxima and minima clusters. The boundary we consider for this method is the zero isocontour of the wind field as any path from a maxima cluster to a minima cluster would certainly pass through this isocontour. Hence for each point on the zero isocontour, we look for its closest local maximum and closest local minimum points. We then consider the maxima cluster and minima cluster to which these local maximum and local minimum points belong, to be associated. A graph that stores all such associations

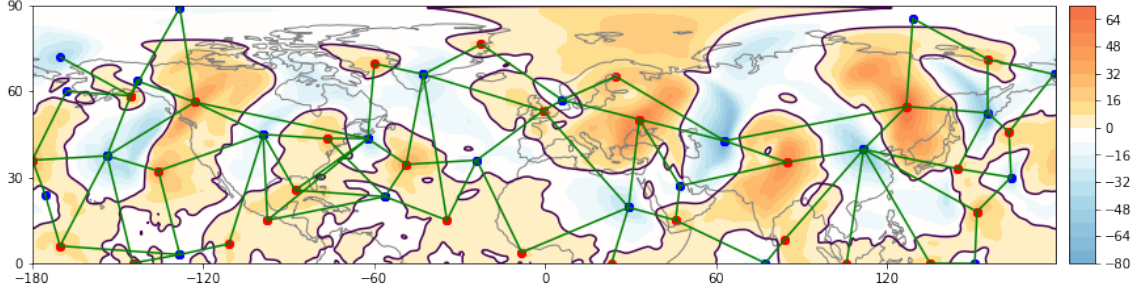


Figure 2.4: Association graph obtained for 300hPa meridional wind field on 21 Jan 2007 (0600 UTC). The blue points represent minima clusters and the red points represent maxima clusters; The solid purple line represents the zero-isocontour; The green line represents associations between maxima and minima clusters.

between maxima and minima clusters is created and is called the association graph. It is a bipartite graph between the maxima and minima clusters whose edges denote associations between maxima clusters and minima clusters. In this graph, a maxima cluster can have more than one minima cluster associated with it and vice versa.

Figure 2.4 illustrates the association graph. For display purposes, we use the point with the largest magnitude of v for each cluster as a representative point for that cluster. The blue points represent the corresponding minima clusters and the red points represent the corresponding maxima clusters; The solid purple line represents the zero-isocontour.

- **Pruning the association graph:** The association graph constructed in the last step may contain several irrelevant edges, i.e, weak associations between maxima and minima clusters that actually belong to different RWPs. Hence, it is necessary to prune the association graph to make sure that individual RWPs are separate and retain only valid associations. To prune the association graph, we assign two types of weights to each edge in the graph: the scalar weight and the estimated gradient. The scalar weight is based on the largest absolute value of v of the component nodes. We can use this to identify the clusters that are part of an RWP by choosing a high threshold value for the scalar weights. The estimated gradient on the other is a more physically motivated measure. RWPs are usually separated by low curvature vorticity areas. The estimated gradient is used to estimate the curvature vorticity linked to a (maxima cluster, minima cluster) pair. Based on our assumption we can use a high enough threshold for the estimated gradient to further prune the association graph. Hence, these two measures

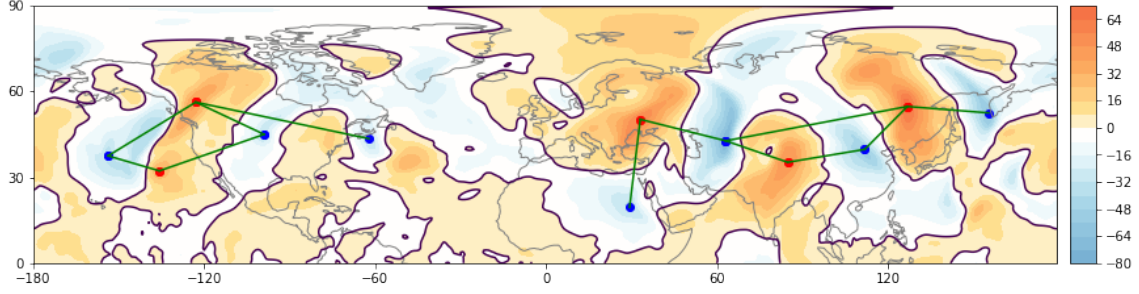


Figure 2.5: Scalar Pruned association graph obtained for 300hPa meridional wind field on 21 Jan 2007 (0600 UTC)

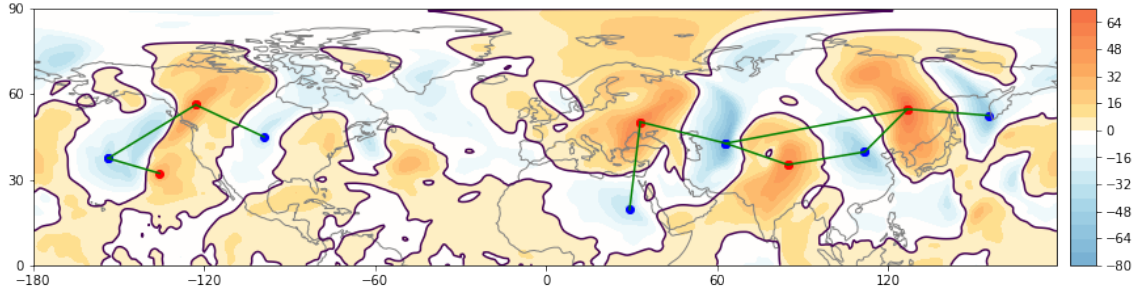


Figure 2.6: Edge Pruned association graph obtained for 300hPa meridional wind field on 21 Jan 2007 (0600 UTC)

together act as a good way to prune the association graph into separate connected components each of which represents regions of RWP activity.

Figure 2.5 shows the association graph after the scalar pruning process. The scalar threshold used in this case is 30m/s. Figure 2.6 shows the graph obtained post both scalar and edge pruning steps. The threshold for estimated gradient used in this case is 0.02. As in the association graph, the blue points represent the corresponding minima clusters and the red points represent the corresponding maxima clusters; The solid black line represents the zero-isocontour.

- Extracting and displaying representative paths: The last step is to extract clear representative paths for each identified RWP from the pruned association graph. This step is purely for visualisation purposes and all information regarding the local maxima and local minima in each cluster is retained. We approach this step as an optimisation problem. The pruned association graph consists of nodes that represent maxima and minima clusters and weighted edges that represent associations and their strength be-

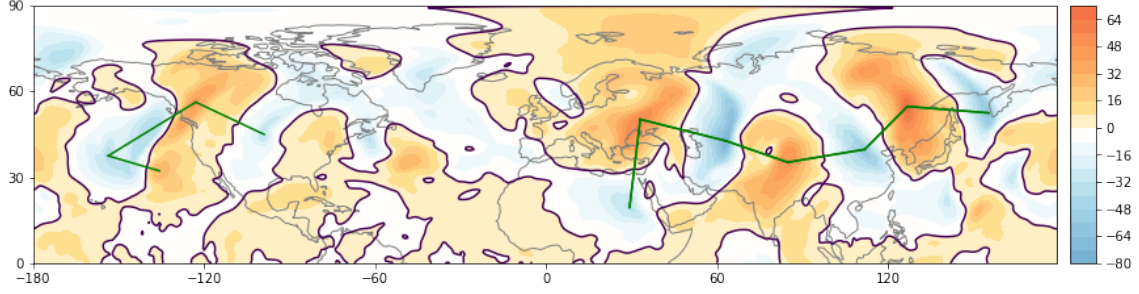


Figure 2.7: Final RWPs obtained for 300hPa meridional wind field on 21 Jan 2007 (0600 UTC)

tween two clusters. For the optimisation, we first compute all the hamiltonian paths within each individual RWP region, i.e, simple paths passing through each node of the graph exactly once. We then assign a path score to each of these paths by summing the edge weights (sum of scalar weight and estimated gradient) of each of the edges constituting the path. Finally, for each RWP, the path with the largest score is extracted and shown as the representative RWP.

Figure 2.7 shows the representative paths obtained for each identified RWP on 21 January 2007 (0600 UTC). Each node represents the local maxima or local minima cluster that constitutes the particular RWP. Each node in this final graph contains all the information of the critical points in that particular cluster. This information gives important geometric properties of an identified RWP.

2.4 Modifications

In this section, we discuss the changes made in the implementation of the above algorithm.

- **Identification of Critical Points:** Critical points of the dataset were located using a neighbourhood based computation in the previous implementation. This was done by specifying a radius and locating a maximum and minimum for areas of this radius. However, this method is dependent on the resolution of the input data and would need revising of the radius parameter with varying resolution of the dataset. To overcome this limitation, we have implemented a grid-based comparison method for determining

the critical points. In this implementation, we compare the scalar value at each grid point to that of its neighbours to determine the local maxima and minima.

- Making use of Pyvista objects: We have used the PyVista module to store the data as opposed to the previous implementation of the algorithm which mainly used `vtk` objects for the same. PyVista is an assistance module for the Visualization Toolkit (VTK) that uses `NumPy` and direct array access to interact with VTK in a different way. This tool provides a Pythonic, well-documented interface for quickly prototyping, analysing, and visualising geographically related datasets. Since all `pyvista` mesh objects are subclasses of the corresponding `vtk` objects, these can be transparently used by `vtk` functions. We use the `pyvista Rectilinear Grid` object to store the data. The Rectilinear Grid can be created using the `numpy` arrays that represent latitude and longitude using the function `pyvista.RectilinearGrid(*args)` (whose arguments are (lon,lat) in this case). We can add the "v" values and the local maxima or local minima values identified by the `addMaxData` or `addMinData` functions respectively (both of which are `numpy` arrays) to the Rectilinear Grid by using the `point arrays` attribute of PyVista gridded objects. The array-like nature of PyVista objects makes it easier to use as opposed to `vtk` objects. This implementation also gets rid of the `paraview` dependency of the algorithm for visualisation and conversion of the dataset from NetCDF files to `vtk` objects. Paraview is a platform for interactive data analysis and scientific visualisation. In the current implementation, we use `xarray`, a python module, that is designed specifically to work with NetCDF files for the conversion of NetCDF files into `numpy` like arrays which can be then used by PyVista to create PyVista objects.
- Changes to the clustering code: As discussed earlier, the method uses the affinity propagation clustering algorithm to partition the identified local maxima and minima into clusters. We choose this algorithm mainly as it supports different similarity measures and also due to its capability to select the number of clusters on its own. In the affinity propagation algorithm, the diagonal weights in the similarity matrix control the number of clusters chosen. If a value close to the minimum possible similarity is chosen, the algorithm will produce a fewer number of clusters, while if a value greater than the maximum possible similarity is chosen, the algorithm will produce a higher number of clusters. We have mainly used this diagonal similarity parameter to optimise the clustering process across several case studies. We chose the value 150 for the diagonal similarity parameter as it gave the most intuitive clusters for all the case studies con-

sidered.

Figure 2.8 shows the output of the clustering algorithm for four different diagonal weights for the wind field on 21 January 2007 (0600 UTC). As mentioned earlier, the red shapes represent minima clusters and the green shapes represent maxima clusters. The black points represent the identified critical points after filtration. As we can see smaller diagonal weights such as 10 and 100 contains a relatively larger number of single-point clusters while higher diagonal weights such as 400 produces fewer clusters but also clusters spatially distant points together.

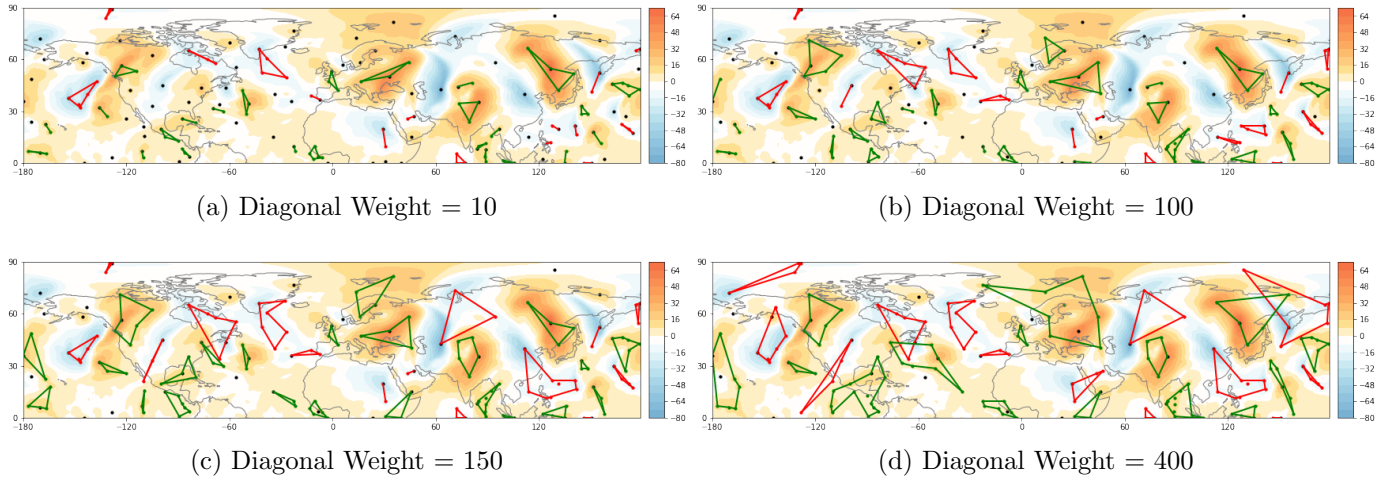


Figure 2.8: Clustering using different diagonal weights in the affinity propagation clustering algorithm. Clusters are computed for meridional wind field at 300hPa (0600 UTC 21 January 2007)

- Changing the association and pruned association graph implementations: In the earlier implementation, the association and pruned association graphs were stored as `vtk` objects. In the modification implemented, all the graphs are stored as `NetworkX` graph objects which are easier to handle than `vtk` objects. These `NetworkX` graph objects consist of nodes that correspond to a maxima or minima cluster and edges between these nodes, which also has weights associated with them. Each node in a graph stores all the information about the constituent points of the corresponding cluster, such as its cluster id, coordinates and scalar value. This information could be useful for further analysis of the extent, amplitude, orientation etc. of the RWP components. A representative point (the point whose absolute scalar value is maximum) is also chosen

for each cluster.

- Changing the representative paths graph computation: As we have seen earlier, after the pruning step, we choose a representative path for each connected component for visualisation purposes. Earlier, this was done by taking all the points in each cluster into consideration. For each connected component, we would look at all the possible paths. The nodes constituting these paths are maxima and minima clusters. All possible combinations between the points in each constituent node of the path are considered, and a path weight for each combination is calculated. These path weights would then be used for choosing the optimal representative paths. In the modification implemented, instead of looking at all possible combinations of the points in each cluster and computing path weights for these, we compute the weight of a path using the edge weights between the nodes that have been previously computed and used for the pruning step. Hence we are avoiding the computation of path weight for each path in the Cartesian product of the critical points in the constituent clusters which takes $O[n^n]$ time. Instead, we directly use the summation of the previously calculated edge weights between the nodes and optimise it over all simple paths within each RWP. This implementation uses far fewer steps as compared to the previous and hence is computationally less complex. Since this step is only for visualisation purposes, we do not lose any valuable information due to this simplification.

Chapter 3

Tracking of RWPs

In this chapter, we discuss the formulation and implementation of the tracking algorithm.

3.1 The Tracking problem

As discussed earlier, RWPs are essential in weather prediction since they have been linked to downstream extreme weather occurrences as well as the development of forecast errors. Tracking RWPs could also help us better understand how the large-scale flow regime is changing. Hence we are interested in exploring an automated feature-based RWP tracking approach similar to [SCC14b]. A feature-based method can substantially simplify the task of efficiently investigating big datasets. Each domain has a unique set of key features. Since these features appear in consecutive time steps, tracking them can improve visualisation and present a useful analytical tool for studying the behaviour of these features. Furthermore, tracking can aid in the automation of searches for events and interactions. It's challenging to keep track of features automatically because they're always changing and interacting. The first attempts of feature tracking [SSZC94] were presented, and the tracking was done using approximations to features. In this work, we describe a new method for tracking that uses a spatial overlap-based approach to correlate features from one dataset to features in a subsequent dataset similar to the volume-based tracking approach presented in [SW96].

3.2 Methodology

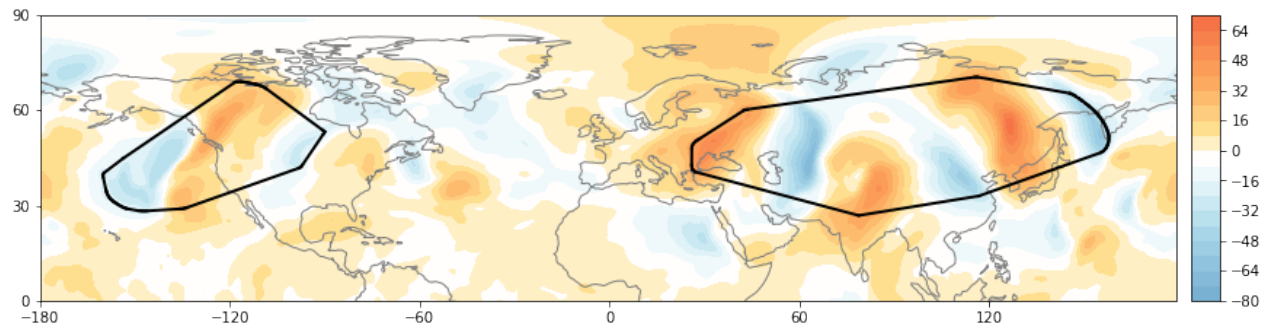
3.2.1 Feature Identification

The first step of a feature-based tracking approach is feature selection. We have already defined the method to identify and store RWPs for any dataset in a graphical form. The regions encompassing each RWPs are the features that we aim to track across time. For this, we first draw closed contours enclosing just one maxima or one minima cluster for each cluster constituting the RWP. We then construct a convex hull encompassing all the regions associated with each extrema cluster constituting the RWP. Hence the region associated with each RWP in a timestep is represented using convex hull objects. All the grid points within that object are attributed to the particular RWP. To construct the contours, we use a factor α of the maximum absolute value in each RWP. To make further processing of these feature geometries simpler, we rasterize the plot to obtain an image array with the input geometries burned in. Rasterisation is the process of converting an image into a collection of pixels, dots, or lines, that, when displayed together, form the image that was previously represented by shapes. This is known as a raster image. Since the raster images obtained are in the form of NumPy arrays, they can be easily used for all further computations.

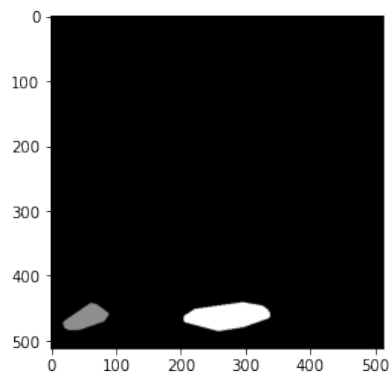
Figure 3.1 shows how the raster image for the features identified for a particular timestep (here, 0600 UTC 21 January 2007) looks like. The polygons in solid black lines in (a) represent the regions associated with each identified RWP in the timestep. The raster image (b) has a dimension of 512 x 512. This is done as we require an image of dimension of the form $2^n \times 2^n$, where n is any natural number, for further processing. Furthermore, the features in the raster image are contained within the original dimension of 90 x 360.

3.2.2 Feature Extraction

Now that we have identified the features to be tracked and stored them in an easily usable form, we need to extract these features to effectively apply spatial overlap techniques. The extraction technique should be broad enough to support a wide range of feature shapes while yet being simple enough to track and quantify. For this purpose, we use quadtrees to extract features in each timestep. Quadtrees are tree data structures in which every node except



(a) Identified features



(b) Raster image

Figure 3.1: (a) Regions of RWP activity (features) identified for 21 January 2007 (0600 UTC) (b) Rasterised image with features identified for 21 January 2007 (0600 UTC) burned in

the root nodes has exactly four children. The quadtree data structure has the advantage of having simple and efficient set operations (union, intersection, and difference calculations, adjacency, membership testing, and transformations such as translation, rotation, and scaling). This efficiency can be attributed to the spatial indexing feature of the quadtrees. We now look at how a quadtree is constructed.

Quadtree construction

A quadtree can be used to represent a $2^n \times 2^n$ image where each pixel corresponds to a feature or the background (denoted by the value 0). Hence, prior to using the quadtree construction algorithm, we convert the raster image to an array of dimension $2^n \times 2^n$. This is done by simply denoting the additional pixels required to be background pixels (of value 0) and would not alter the computation.

To construct the quadtree, we first create and insert the root node that represents the entire region covered by the image (it consists of all the features present in that particular timestep). Then the image is subdivided into four regions. If any of the subdivisions do not entirely correspond to a single feature or the background, then it is again subdivided into 4. This process of subdividing is continued recursively until all leaf nodes of the quadtree either represent a region containing a single feature or a region containing only the background. Each of the nodes contains information about all the features enclosed by the particular region. It is constructed in the form of a directed graph where edges are directed from the parent node to the daughter node. The quadtree is divided into different levels and the nodes corresponding to each level denote regions of a particular size. Also, to facilitate spatial indexing, we index the daughter nodes of a node i as $(4i + j)$ where j is a number between 1 and 4 (both included). This is useful as for two images of the same dimension, the nodes with the same index denote the same region in the images, spatially.

It is apparent from this formulation that quadtrees save a lot of space while storing images. Rather than storing a large 2-D array of every pixel in the image, a quadtree can record the same information at many divisive levels higher than the pixel-resolution-sized cells we'd otherwise need. The height and resolution of the tree are limited by the pixel and image sizes. The algorithm also returns the number of pixels corresponding to each feature in the timestep.

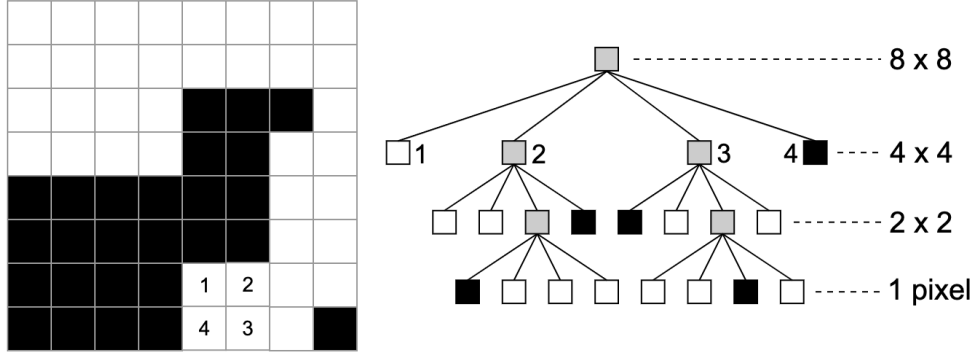


Figure 3.2: Quadtree representation of an image of dimension 8x8[Mu8]

Figure 3.2 shows how an image of size 8x8 is represented by a quadtree. The black pixels in the image represent features and the white pixels represent the background. Similarly, the white nodes in the quadtree represent a spatial region that is entirely part of the background and black nodes represent a region that is entirely constituted by feature pixels. Grey nodes indicate a region which consist of both feature and background pixels.

3.2.3 Feature Tracking

Once we have defined and extracted the features for each timestep, we can define an algorithm to track these features across time. Prior to that, we discuss a few evolutionary events that occur in time-varying datasets.

- Continuation is an event in which a feature in the current timestep continues in the next timestep. The feature may undergo rotation or translation and may even strengthen (become bigger- grow) or weaken (become smaller) in size.

Figure 3.3 (a) shows an example of continuation of features from one timestep to the next. An RWP region identified on 26 January 2007 (0600 UTC) overlaps with an RWP region identified on 27 January 2007 (0600 UTC). The polygon in solid black line represents the RWP region identified on 27 January 2007 and the polygon in the dotted black line represents the RWP region identified on 26 January 2007.

- Creation is an event where a feature absent in the current timestep emerges in the next

timestep.

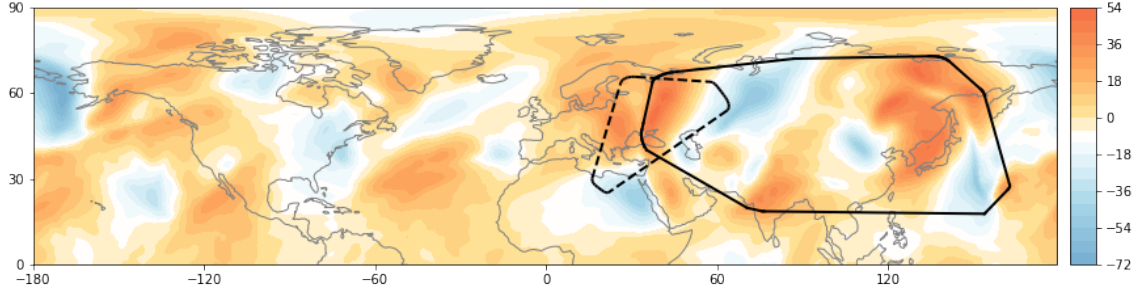
- Dissipation is an event where a feature present in the current timestep weakens and becomes part of the background in the next timestep.
- Bifurcation is an event where a feature in the current timestep splits into two or more features in the next timestep.

Figure 3.3 (b) shows an example of features that split in consecutive timesteps. An RWP region identified on 21 January 2007 (0600 UTC) overlaps with two RWP regions identified on 22 January 2007 (0600 UTC). The polygons in solid black line represent the RWP regions identified on 22 January 2007 and the polygon in dotted black line represents the RWP region identified on 21 January 2007.

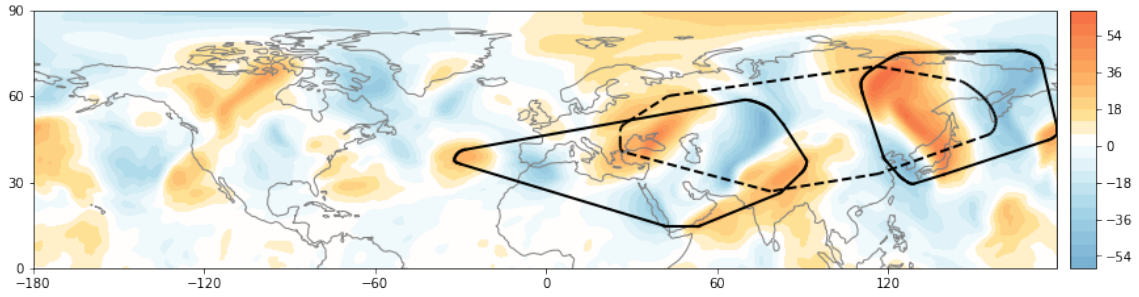
- Amalgamation is an event where two or more features in the current timestep merge into a single feature in the next timestep.

Figure 3.3 (c) shows an example of features that merge in consecutive timesteps. Two RWP regions identified on 22 January 2007 (0600 UTC) overlaps with one RWP region identified on 23 January 2007 (0600 UTC). The polygon in solid black line represents the RWP region identified on 23 January 2007 and the polygons in dotted black line represent the RWP regions identified on 22 January 2007.

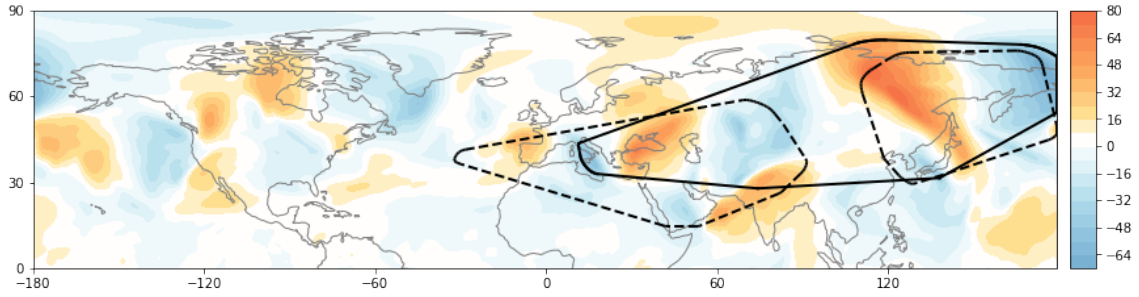
We now discuss the algorithm to compute the spatial overlap between features in two consecutive time steps. The features in each timestep are stored in the form of quadtree graphs. We make use of the spatial indexing feature of the quadtrees for this computation. Consider two quadtrees for consecutive timesteps t_i and t_{i+1} . For each index i , if it is part of both the quadtrees, we compare the nodes at this index as follows: If one of the nodes is a background node, we do not consider the descendants of this node in either tree in further iterations. If both the nodes are feature nodes, the regions corresponding to the index overlap. The extent of this overlap can be computed using the number of pixels the region consists of based on the level (height) of the node in the tree. If both the nodes are mixed nodes (non-leaf nodes), then we further compare the descendants of these nodes corresponding to the same indices. If one of the nodes is a mixed node while the other is a feature node, we traverse the tree rooted at the mixed node and identify the leaf nodes corresponding to individual features. These leaf nodes would represent regions of overlap with the feature from the other timestep.



(a) Continuation of features in consecutive timesteps



(b) Features split in consecutive timesteps



(c) Features merge in consecutive timesteps

Figure 3.3: Evolutionary events that occur in time-varying datasets. Polygons in solid black line represents RWP polygons identified in the current timestep and the polygons in dotted black line represents RWP polygons identified in the previous timestep

The extent of the overlaps can be computed using the number of pixels the regions consist of based on the level (height) of the nodes in the tree.

Chapter 4

Results and Conclusions

4.1 Feature Identification Calculations

As we previously discussed, feature identification is the first step in a feature-based tracking pipeline. The method we implemented chooses regions of high amplitude meridional winds, which encloses the RWPs identified previously, as features. For this we choose a fraction of the absolute maximum scalar value in each identified RWP as the threshold for the construction of level sets. This ensures that the resulting polygons enclosing each RWP contains only regions of significance and the tracking is not much affected by weaker regions (smaller amplitude components) of the identified wave packets. Based on case studies, we have found that using 0.5 as the fraction for computation of thresholds yields reasonable enclosing regions for the identified RWPs. Figure 4.1 shows bounding polygons calculated for 3 factors - 0.25, 0.5 and 0.75 for RWPs identified on 20 January 2007. The bold black line represents the representative wave packets for this timestep. The red shapes represent maxima clusters and the green shapes represent minima clusters.

4.2 Tracking Graph Calculations

For tracking the regions (features) identified, our approach has been to match the regions in consecutive timesteps based on their spatial overlap. We compute the overlap between fea-

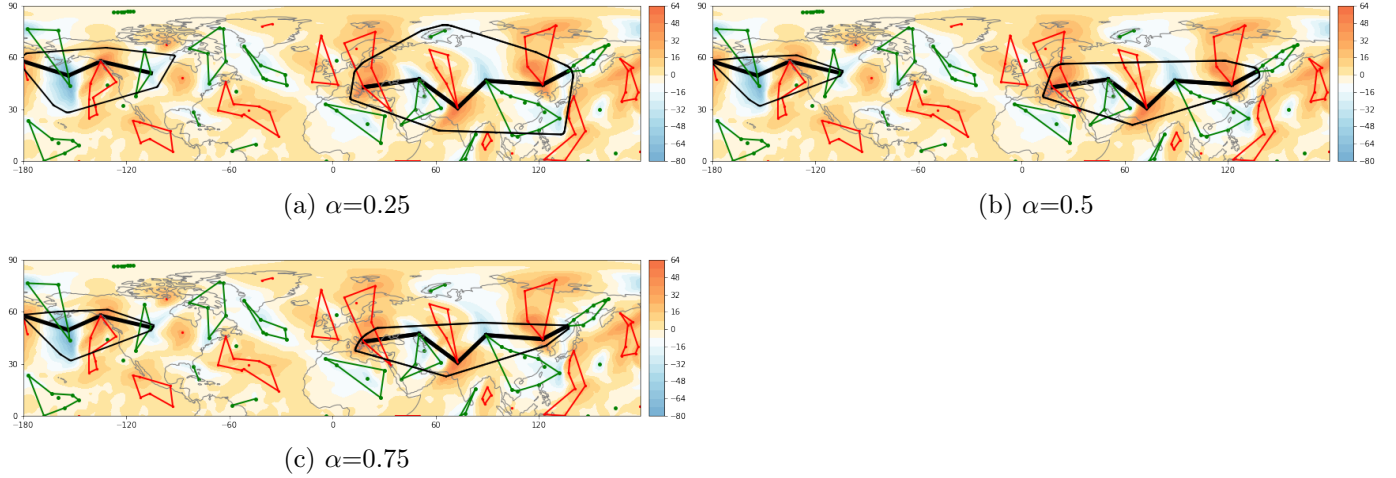


Figure 4.1: Regions identified for individual RWPs using different factors. The panel shows meridional wind field at 300hPa (0600 UTC 20 January 2007). The bold black lines denote the representative RWPs in that timestep and the polygons in the thin black lines denote the identified feature

tures in two adjacent timesteps by performing an intersection operation using the quadrees that store the features for each time step. This operation gives the result in the form of the number of pixels overlapped between each pair of overlapping features. This can be represented in the form of a multipartite graph where each independent set of the vertices of this graph represent a particular timestep and the nodes in this set represent the features of this timestep. The edges of this graph would represent extend of overlap between the respective features.

As we saw for the computation of the association graph in the identification pipeline, each overlap detected by this algorithm may not be significant and represent the evolution of RWPs across time. Hence we assign a measure of significance to each overlap detected. The measure that we use for this implementation (similar to that used by [SW96]) is given below:

$$W(A^i, B^{i+1}) = \frac{Q(A^i, B^{i+1})}{\max(Q_A^i, Q_B^{i+1})}$$

where A is a feature in timestep t_i and B is a feature in timestep t_{i+1} ; $Q(A^i, B^{i+1})$ denotes the number of pixels that constitute the region of overlap between A and B ; Q_A^i represents the total number of pixels that constitute the feature A and Q_B^{i+1} represents the total number of pixels that constitute the feature B .

Pruning the overlaps detected between features of consecutive timesteps based on this measure using an appropriate threshold value would help get rid of irrelevant overlaps. This measure is also useful for the computation of bifurcations and amalgamations of features. In these cases, we use the combinations instead of single features in the weight calculation. For example, Let's consider the case where a feature A in timestep t_i overlaps with features B and C in timestep t_{i+1} . We can compute the weight for all three cases- $W(A^i, B^{i+1})$, $W(A^i, C^{i+1})$ and $W(A^i, (B \cup C)^{i+1})$. The combination with the maximum weight is chosen as the track. Since we use a quadtree representation for the features, computation of all the parameters involved in weight calculation can be done simultaneously.

Using the above measure, we have computed the weights for all possible combinations of overlapping features (including unions) and retained the combination with the maximum weight. Furthermore, we could also prune these edges using a threshold for the edge weights to ensure that no irrelevant tracks are being identified.

Figures 4.2 and 4.3 describe the evolution of a feature using the tracking method described above. Here we have represented 4 timesteps in each figure. The tracking is done from 12 January 2007 to 18 January 2007 (Figure 4.2) and 16 January 2007 to 26 January 2007 (Figure 4.3). The polygons using solid black lines represent a region of RWP activity (feature) in the current timestep and the polygons using dotted black lines represent a region of RWP activity (feature) in the previous timestep. Black points denote the centroids (representative points) of the RWP polygons in the current timestep whereas red points denote the centroids (representative points) of the RWP polygons in all the previous timestep starting from 12 January 2007 (for Figure 4.2) and 16 January 2007 (for Figure 4.3). Black solid lines represent the overlap between features of the current timestep and the preceding timestep. The dotted black lines represent all the overlaps detected previously.

As we can see, although the algorithm captures RWP movement more or less accurately, there are some anomalies. For example, from 15 January 2007 to 16 January 2007, the RWP being tracked seems to have moved towards the west (as can be seen from the positions of the polygons). However, this is contrary to what we expect due to the property of downstream development of RWPs. The larger polygon on 16 January encompassing the polygon on 15 January could either be the result of the amplification of the weaker RWP on 15 January,

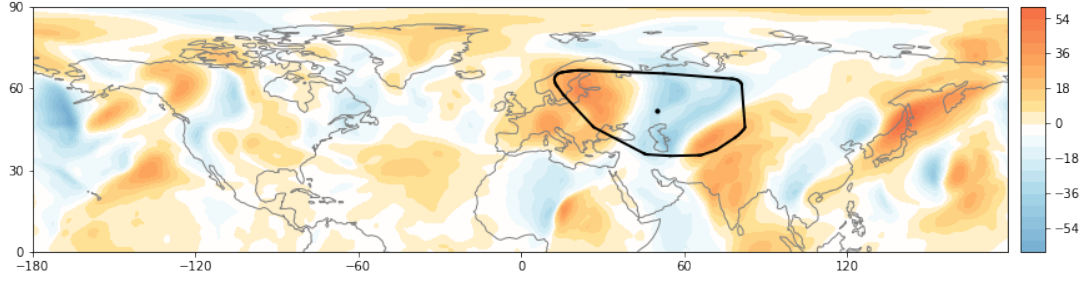
or it could be the result of an amalgamation of the RWP on 15 January with another RWP propagating towards the west. We can also see that the RWP during the time interval from 16 January to 18 January 2007 increases in size towards the eastern end while the western end is almost stationary. This could indicate a split of the initial RWP. However, this split is not being captured by the tracking algorithm. Another anomaly is that in Figure 4.3, we can see that a feature splits into two on 22 January, merges back into one feature on 23 January and splits again into two on 24 January. This should ideally have been a single RWP.

4.3 Discussion and Conclusion

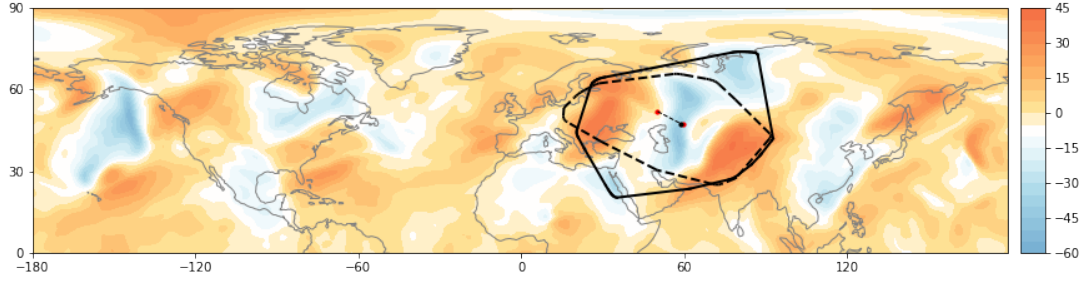
We have made several modifications in the identification algorithm for better computational efficiency. These include: implementing an input dataset resolution-independent computation to locate local maxima and minima; Getting rid of paraview dependency by using simpler alternatives for computation and visualisation such as `Pyvista`, `Numpy`, `xarray`, `Matplotlib` etc; changing the clustering code so as to obtain more intuitive clusters; changes in the graph implementations for easier handling and reduced computational complexity without losing any phase information. This method also facilitates the identification of RWPs of any orientation unlike previous attempts at identification.

We have also implemented an algorithm to track these RWPs. The quadtree representation of features in each timestep optimises the storage of these features. It also facilitates simultaneous and easier computation of overlap between features and the size of these features. We also construct a tracking graph that describes the evolution of RWPs across time which also captures splits and merges of features. The graph representation of tracking makes it easy to analyse and also use filtering techniques.

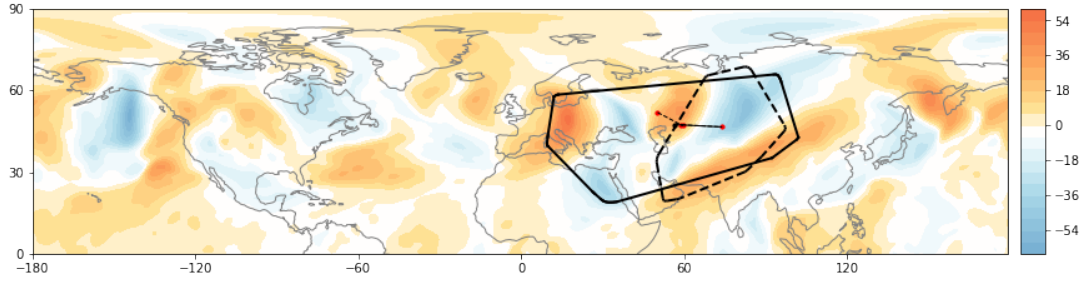
Although conceptually thorough, it has been observed that the tracking algorithm sometimes produce counter intuitive results. Due to the complexity of RWP development, we have not yet been able to get optimal RWP tracks using the tracking algorithm. We would need further validation and modifications to obtain physically intuitive results for tracking. Another limitation of this method is that due to 2D projection of geographic data that we use, we cannot often identify and track every RWP. This could be overcome by making



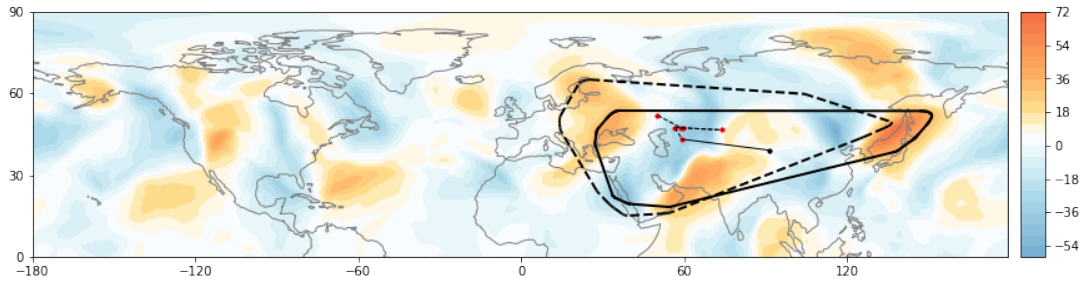
(a) 12 January 2007



(b) 14 January 2007

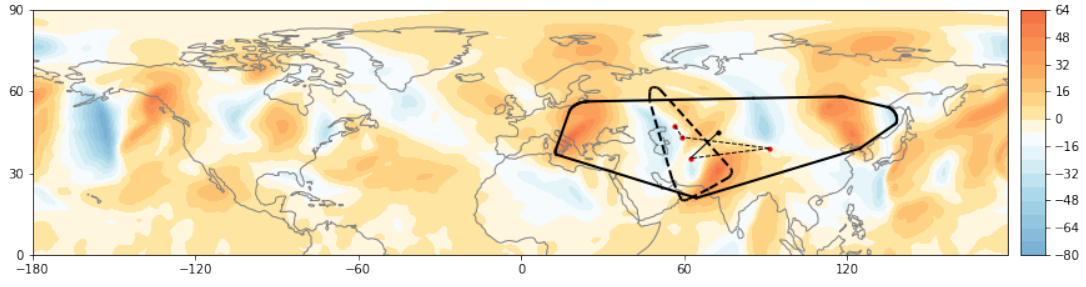


(c) 16 January 2007

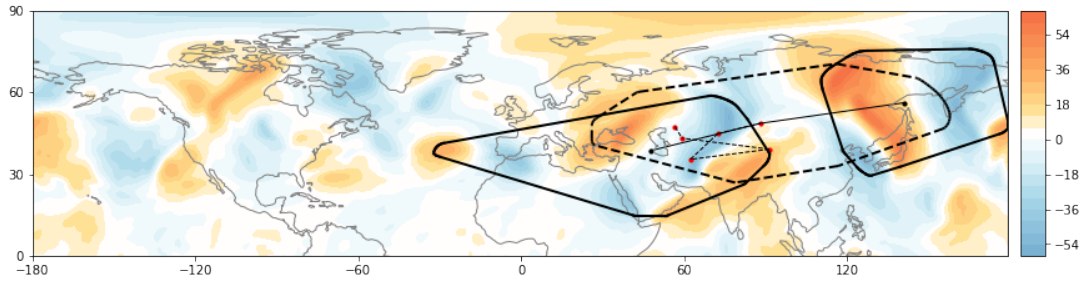


(d) 18 January 2007

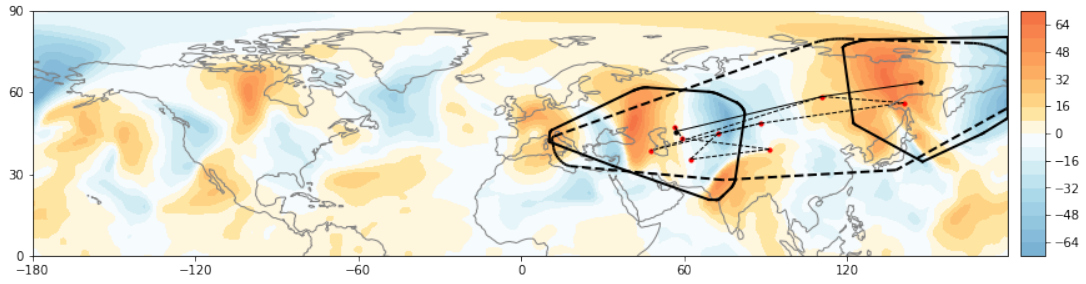
Figure 4.2: Regions identified for individual RWPs using different factors. The panel shows meridional wind field at 300hPa (0600 UTC 12 January - 18 January 2007). Black points represent centroids of features identified in the current timestep. Red points represent centroids of features identified in the previous timesteps. The solid black lines connecting red and black points denote the movement of the RWP from the previous timestep to the current. The dotted black lines connecting red and red points denote all the previous movements of the RWP.



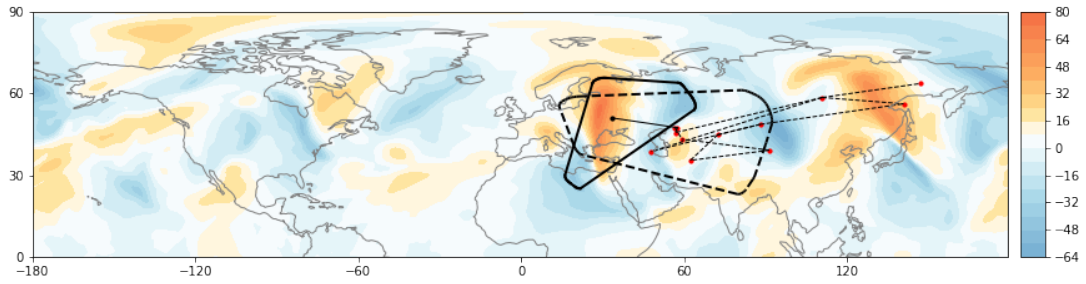
(a) 20 January 2007



(b) 22 January 2007



(c) 24 January 2007



(d) 26 January 2007

Figure 4.3: Regions identified for individual RWPs using different factors. The panel shows meridional wind field at 300hPa (0600 UTC 20 January - 26 January 2007)

corrections for identification and tracking at the edges or using a spherical projection for the dataset. Also, due to the formulation of features as polygons enclosing sublevel and superlevel sets of a particular threshold value, features in the same timestep may overlap.

All the data used in this work can be downloaded from ECMWF. The identification and tracking code will be available shortly open source.

Bibliography

- [ABKC13] Heather M. Archambault, Lance F. Bosart, Daniel Keyser, and Jason M. Cordeira. A climatological analysis of the extratropical flow response to recurving western north pacific tropical cyclones. *Monthly Weather Review*, 141(7):2325 – 2346, 2013.
- [DNN13] Harish Doraiswamy, Vijay Natarajan, and Ravi S. Nanjundiah. An exploration framework to identify and track movement of cloud systems. *IEEE Transactions on Visualization and Computer Graphics*, 19(12):2896–2905, 2013.
- [FD07] Brendan J. Frey and Delbert Dueck. Clustering by passing messages between data points. *Science*, 315(5814):972–976, 2007.
- [GDJ⁺11] Ilona Glatt, Andreas Dornbrack, Sarah Jones, Julia Keller, O. Martius, Aurelia Muller, Dieter H. W. Peters, and Volkmar Wirth. Utility of hovmöller diagrams to diagnose rossby wave trains. *Tellus A: Dynamic Meteorology and Oceanography*, 63(5):991–1006, 2011.
- [GFW18] Paolo Ghinassi, Georgios Fragkoulidis, and Volkmar Wirth. Local finite-amplitude wave activity as a diagnostic for rossby wave packets. *Monthly Weather Review*, 146(12):4099 – 4114, 2018.
- [GV] Federico Grazzini and Frédéric Vitart. Atmospheric predictability and rossby wave packets. *Quarterly Journal of the Royal Meteorological Society*, 141(692):2793–2802.
- [Hak05] Gregory J. Hakim. Vertical structure of midlatitude analysis and forecast errors. *Monthly Weather Review*, 133(3):567 – 578, 2005.
- [HLH⁺16] C. Heine, H. Leitte, M. Hlawitschka, F. Iuricich, L. De Floriani, G. Scheuermann, H. Hagen, and C. Garth. A survey of topology-based methods in visualization. *Computer Graphics Forum*, 35(3):643–667, 2016.
- [KHNH12] Jens Kasten, Ingrid Hotz, Bernd Noack, and Hans-Christian Hege. Vortex Merge Graphs in Two-dimensional Unsteady Flow Fields. In Miriah Meyer and

- Tino Weinkauffs, editors, *EuroVis - Short Papers*. The Eurographics Association, 2012.
- [Mu8] Wojciech Mula. Bitmap and its compressed quadtree representation, 2008.
- [PMN20] Karran Pandey, Joy Merwin Monteiro, and Vijay Natarajan. An integrated geometric and topological approach for the identification and visual analysis of rossby wave packets. *Monthly Weather Review*, 148(8):3139 – 3155, 2020.
- [RKG⁺11] Jan Reininghaus, Natallia Kotava, David Günther, Jens Kasten, and Ingrid Hotz. A scale space based persistence measure for critical points in 2d scalar fields. *IEEE Transactions on Visualization and Computer Graphics*, 17:2045–52, 12 2011.
- [RKWH12] Jan Reininghaus, Jens Kasten, Tino Weinkauff, and Ingrid Hotz. Efficient computation of combinatorial feature flow fields. *IEEE Trans. Vis. Comput. Graph.*, 18(9):1563–1573, September 2012.
- [SCC14a] Matthew B. Souders, Brian A. Colle, and Edmund K. M. Chang. The climatology and characteristics of rossby wave packets using a feature-based tracking technique. *Monthly Weather Review*, 142(10):3528 – 3548, 2014.
- [SCC14b] Matthew B. Souders, Brian. A. Colle, and Edmund K. M. Chang. A description and evaluation of an automated approach for feature-based tracking of rossby wave packets. *Monthly Weather Review*, 142(10):3505 – 3527, 2014.
- [SSZC94] R. Samtaney, D. Silver, N. Zabusky, and J. Cao. Visualizing features and tracking their evolution. *Computer*, 27(7):20–27, 1994.
- [SW96] Deborah Silver and Xin Wang. Volume tracking. *Proceedings of Seventh Annual IEEE Visualization '96*, pages 157–164, 1996.
- [VMN⁺19] Akash Anil Valsangkar, Joy Merwin Monteiro, Vidya Narayanan, Ingrid Hotz, and Vijay Natarajan. An exploratory framework for cyclone identification and tracking. *IEEE Trans. Vis. Comput. Graph.*, 25(3):1460–1473, March 2019.
- [WRCM18] Volkmar Wirth, Michael Riemer, Edmund K. M. Chang, and Olivia Martius. Rossby wave packets on the midlatitude waveguide—a review. *Monthly Weather Review*, 146(7):1965 – 2001, 2018.
- [YMS⁺21] Lin Yan, Talha Bin Masood, Raghavendra Sridharamurthy, Farhan Rasheed, Vijay Natarajan, Ingrid Hotz, and Bei Wang. Scalar field comparison with topological descriptors: Properties and applications for scientific visualization. *Computer Graphics Forum*, 40(3):599–633, 2021.

- [ZSP⁺03] Aleksey V. Zimin, Istvan Szunyogh, D. J. Patil, Brian R. Hunt, and Edward Ott. Extracting envelopes of rossby wave packets. *Monthly Weather Review*, 131(5):1011 – 1017, 2003.